

BTC and ETH Trading Strategies

Inter IIT Bootcamp | Quant '26 | BTC/USDT and ETH/USDT | Jan 2021 - Dec 2025

1. Summary

We built one rule-based breakout strategy per asset. Each uses a daily trend filter, a 4-hour breakout entry, ATR-based stops, volatility-based sizing and drawdown controls. The rules can be inspected trade by trade. Before settling on them we tried several other approaches (statistical models, mean reversion, correlation, machine learning) and reviewed four published papers. Section 4 explains why none of those were used. Section 6 tests how the results change when the parameters are varied.

Mode / asset	CAGR	Sharpe	Max DD	Quarter wins
BTC long + short	52.81%	1.72	23.59%	11 / 20
BTC long only	41.91%	1.54	23.59%	11 / 20
ETH long + short	65.86%	1.46	24.51%	10 / 20
ETH long only	42.51%	1.16	28.69%	10 / 20

BTC beats the "more than half of quarters" target. ETH reaches exactly half, so that target is not met.

2. Data and Accounting

- 175,296 UTC bars per asset (15-minute), 1 Jan 2021 - 31 Dec 2025. Each account starts with 100,000 USDT.
- Data checks: duplicates, bad prices, invalid OHLC and gaps are rejected. 80 zero-volume bars per file cannot trade.
- Signals use only **completed** 4-hour and daily candles. Entries fill at the next tradable 15-minute open. Stops can fill inside a bar; an adverse gap fills at the open.
- Costs: 0.15% of notional per transaction (fee plus slippage), and 5% annual carry on shorts. Unused cash earns zero.
- The final position is closed at the last open. Buy-and-hold is costed the same way.
- Sharpe uses daily returns, zero risk-free rate, $\sqrt{365}$. Max drawdown uses 15-minute marks. A "quarter win" means the account beat costed buy-and-hold in that quarter.
- The engine is custom NumPy/pandas code, with no backtesting library.

3. Final Strategies

BTC (slower, more selective)

- Long: daily close above 40-day SMA; 4h close above previous 36-bar high; volatility rank ≤ 0.90 ; weekdays only. Stop 4 ATR; profit lock after +1.25 ATR; 54-bar exit channel.
- Short: daily close below 99% of 75-day SMA, with the SMA lower than 30 days earlier; break of previous 72-bar low; Hurst ≥ 0.45 . Stop 3 ATR; lock at +1.5 ATR; 4 ATR trail; exit on 18-bar channel or loss of bearish regime.
- Risk: long cap 100%, short cap 80%; drawdown thresholds 15% / 26%; 1-day cooldown; 60-day max hold.

ETH (faster)

- Long: daily close above 35-day SMA; 4h close above previous 10-bar high; ADX ≥ 20 ; volatility rank ≤ 0.90 ; weekends allowed. Stop 2 ATR; lock after +2.5 ATR; 36-bar exit channel.
- Short: daily close below 35-day SMA (itself below its value 10 days earlier); 4h close below previous 36-bar low. Stop 2.5 ATR; 18-bar exit channel.
- Risk: long cap 100%, short cap 65%; drawdown thresholds 25% / 30%; 30-minute cooldown; 60-day max hold.

The BTC short rules were redesigned after trade-level review. Short P&L went from 6,744 to 146,416 USDT. Three trades still give about 82% of BTC short profit, so the result is concentrated trend capture rather than steady income.

4. Strategies We Tried That Did Not Work, and Why

4A. Our own experiments

#	Strategy	Idea	Why it did not work
1	Indicator-based statistical strategy (RSI, MACD, moving-average crossovers)	Buy and sell on standard indicator signals.	Crypto trends in sudden bursts with long quiet periods. Fixed indicator thresholds give many false signals in the quiet periods. Each small loss pays 0.30% round-trip cost, and the few big wins do not cover them.
2	Auto-regression (AR)	Predict the next return from the past few returns.	Crypto returns have almost no linear memory. The coefficients come out near zero and change from one period to the next. A forecast this weak is smaller than the trading cost.
3	Ornstein-Uhlenbeck (mean reversion)	Assume price returns to an average, so buy dips and sell rallies.	This only holds in sideways markets. BTC and ETH spend long stretches in strong trends (2021, 2024). In a trend, "cheap" gets cheaper and the strategy keeps buying into a fall. The fitted reversion speed also drifts over time, so the model is wrong when it matters.
4	Cross-sectional correlation strategy	Pick the asset that moves most early in the session and trade the relationship with others.	We trade only two highly correlated coins (BTC and ETH). With so few assets there is little spread to rank, and in stress both fall together. The "first-hour mover" signal is noisy for a market that trades 24/7 and has no real opening.
5	Machine learning: Random Forest on three return signals (intraday return, return vs previous close, return vs open)	Let the model learn which return patterns come before up-moves.	(a) It ran very slowly on 175,000+ bars per asset. Tuning, retraining by time window and repeating tests was too slow to finish in the time available, which risks losing points. (b) The three inputs are built from the same prices and carry nearly the same information. (c) A model can be right slightly more than 50% of the time and still lose money after costs. (d) A forest trained on one market phase tends to fail in the next. (e) A forest is hard to explain trade by trade. We chose rules we can inspect.

4B. Published research we reviewed, and why we did not use it

BTC papers

(1) Garcia & Schweitzer (2015), "Social signals and algorithmic trading of Bitcoin."

It trades on Twitter valence, opinion polarization and exchange volume.

- *Data we do not have.* It needs tweets, search trends and exchange-wide volume. The competition gives only BTC and ETH 15-minute price/volume data.
- *Market has changed.* The test window was a falling market (2014) with a small, early-stage Bitcoin market. The authors themselves say profits may vanish once other traders use the same signals.
- *Costs.* Their own cost test shows the strategy stops being profitable above about 0.25% cost. Our assumed 0.15% per trade plus carry leaves little room for a strategy that trades every day.
- *Short-selling assumption.* It uses fully invested daily trades, with shorts closed at once. This is not realistic for our drawdown-controlled account.

(2) Otabek & Choi (2024), "Multi-level deep Q-networks for Bitcoin trading strategies."

It uses three stacked deep-reinforcement-learning models plus Twitter sentiment.

- *Very short test.* The headline 29.93% return and Sharpe near 2.7 come from a test of only 30 days (720 hours). That is far too short to show a reliable edge, and ours covers five years.
- *Fee level.* It assumes a 1.5% fee, and the return is reported over that short test, not as a multi-year CAGR. It cannot be compared with our figures.
- *Heavy and slow.* Three neural networks (one with 20,001 outputs) need long training and many runs with random starts. In a competition with time limits, this is slow and risks lost points.

- *Needs sentiment data* we do not have, and the reward function is hand-built, which makes results sensitive to its settings.
- *Not transparent.* We cannot easily say why it bought or sold a given trade.

ETH papers

(3) Yeganeh et al. (2025), "Attention layer ... ratio control chart ... Ethereum."

It watches the ratio of the 50- and 100-hour moving averages and trades when a control chart signals.

- *Weak backtest.* It uses \$100 of capital, long-only trades, and about 13 months of hourly data (Nov 2022 – Dec 2023). That window was mostly a rising market, so it flatters a long-only method.
- *Tuned on the same data.* The best setting (ratio 0.5) was chosen by looking at the same backtest, so the result is likely over-fitted.
- *Small edge.* The reported return is about \$23 on \$100 over the whole period. With wins and losses this small, real fees and slippage could remove most of it.
- *No short side or risk limits.* No position sizing, stops or drawdown controls, which our brief requires.
- *Slow to design.* Choosing its parameters needs repeated Monte Carlo simulation.

(4) Sebastião & Godinho (2021), "Forecasting and trading cryptocurrencies with machine learning under changing market conditions."

It uses linear models, random forests and SVMs on daily data, combined into ensembles.

- *Daily data only,* from 2015–2019. It cannot show how a method works on 15-minute bars in 2021–2025.
- *Poor individual models.* The paper reports that five of its 18 models were right less than half the time on the test period, and several were worse than a naive guess.
- *Costs destroy the edge.* After a 0.5% round-trip cost, its Bitcoin ensembles lost money or earned about 1%. Its best result was about 9.6% a year (ETH), after costs, in a falling market.
- *Long-only, and trades rarely.* The best ensemble was invested only 9–17% of days, so it cannot capture the big trends that give our strategy most of its gains.
- *Models need retraining.* A rolling retrain of many models is slow to run.

4C. Design changes we tested and rejected

Besides trying other models, we tested changes to our own breakout strategy. For each one we show what we changed, what happened, why it failed, and what we did instead. For comparison, the final strategies reach 52.81% CAGR (BTC) and 65.86% CAGR (ETH).

1. Funded BTC ensemble (blending several strategies)

- *What we tried:* split the BTC account across several smaller strategies and combine their signals.
- *What happened:* 23.84% CAGR and 1.19 Sharpe, with the account invested only about 19% of the time.
- *Why it failed:* each strategy held only a slice of the money, so even good signals made little difference. Thousands of small orders each paid the 0.30% round-trip cost, and the result was hard to explain trade by trade.
- *What we did instead:* one breakout process that puts the full allowed size behind a single clear signal.

2. Copy the ETH short rules to BTC

- *What we tried:* use the same short rule on BTC that works on ETH.
- *What happened:* the standalone BTC short account lost 9.26%.
- *Why it failed:* the ETH rule reacts quickly (35-day average, 36-bar low). BTC often bounces sharply inside a downtrend, so the quick rule kept shorting small dips and getting stopped out. The two coins look alike on a chart, but their moves behave differently.
- *What we did instead:* a slower BTC short rule. It needs a longer downtrend (price below 99% of the 75-day average, with that average falling over 30 days), a break of the previous 72-bar low, and a trend-strength check ($Hurst \geq 0.45$).

3. Only widen the BTC short stop

- *What we tried:* keep the old short entries but give each short more room before it is stopped out.
- *What happened:* 3.41% short CAGR with a 28.62% drawdown.
- *Why it failed:* wider stops help good trades survive, but they do not fix bad entries. A bad entry now simply loses more before it is closed.
- *What we did instead:* fixed the entry rule as well as the exit. This lifted BTC short profit from 6,744 to 146,416 USDT.

4. Block all ETH trades after a loss

- *What we tried:* after a losing trade, stop trading both long and short for a while.
- *What happened:* ETH CAGR fell to 51.77%.
- *Why it failed:* ETH often reverses quickly. Blocking both directions made us miss the recovery trade that follows many losses.
- *What we did instead:* a short 30-minute cooldown after a normal stop-out. A three-day pause is used only when the drawdown limit is hit.

4D. What we learned

1. Costs of 0.30% round-trip kill any strategy that trades often with a small edge.
2. Crypto spends long periods trending, so mean reversion fails, and linear prediction has almost no signal.
3. Results from short or favourable test windows (30 days, one rising year) do not carry over to five years.
4. Sentiment and social-media data are not available in this competition.
5. Complex models run slowly and are hard to explain. Simple rules with clear stops can be checked for every trade.

5. Robustness Tests (Monte Carlo and HMM)

Strategies were replayed on 1,000 synthetic five-year paths per asset and method, with costs and carry.

Asset / test	Median CAGR	Median Sharpe	DD95	Profitable paths
BTC / Block MC	29.73%	1.18	35.93%	96.6%
BTC / HMM	6.70%	0.42	39.28%	75.2%
BTC / Fair walk	-4.41%	-0.21	51.88%	26.5%
ETH / Block MC	31.41%	0.93	40.47%	95.2%
ETH / HMM	-6.32%	-0.24	61.83%	19.2%
ETH / Fair walk	-6.35%	-0.22	63.87%	21.1%

The strategies hold up when real trend patterns are kept (block MC). They struggle when trend order is scrambled (HMM, fair walk), which confirms they depend on trends. This supports conditional resilience, not universal robustness.

6. Parameter Sensitivity (100 Variations per Asset)

To check that the results do not rest on one exact parameter set, each strategy (long + short) was re-run on the same data, costs and carry with 100 variations. Every variation moves all of the parameters below at once, each within a narrow band around its final value. Variations were drawn with a fixed seed (26), contain no duplicates, and are not filtered by outcome. Code: sweep.py.

Parameter	BTC range tested (final)	ETH range tested (final)
Daily trend SMA (days)	30 – 50 (40)	30 – 45 (35)
Long breakout (4h bars)	28 – 44 (36)	8 – 14 (10)
Long exit channel (bars)	42 – 66 (54)	30 – 42 (36)
Long stop (ATR)	3.5 – 4.5 (4.0)	1.75 – 2.5 (2.0)

Parameter	BTC range tested (final)	ETH range tested (final)
Long profit lock (ATR)	1.0 – 1.5 (1.25)	2.0 – 3.0 (2.5)
Long trailing stop (ATR)	fixed	5 – 7 (6)
Volatility rank cap	0.85 – 0.95 (0.90)	0.85 – 0.95 (0.90)
ADX minimum (long)	fixed	15 – 25 (20)
Position volatility target	0.7 – 0.9 (0.8)	fixed
Short SMA (days) / slope window	65 – 85 (75) / fixed	shared SMA / 8 – 12 (10)
Short breakout (bars)	60 – 84 (72)	30 – 42 (36)
Short exit channel (bars)	14 – 22 (18)	14 – 22 (18)
Short stop (ATR)	2.5 – 3.5 (3.0)	2.25 – 2.75 (2.5)
Short profit lock (ATR)	1.25 – 1.75 (1.5)	1.25 – 1.75 (1.5)
Short trailing stop (ATR)	3.5 – 4.5 (4.0)	3.5 – 4.5 (4.0)
Hurst minimum (short)	0.40 – 0.50 (0.45)	fixed
Short cap	70% – 90% (80%)	55% – 75% (65%)
Drawdown thresholds	12/24, 15/26, 18/28 (15/26)	22/28, 25/30, 28/33 (25/30)
Cooldown (15-min bars)	fixed	1 – 4 (2)

Table 6.1 shows the median of each metric across the parameter variations, next to the final strategy.

Median metric	BTC final	BTC median of parameter variations	ETH final	ETH median of parameter variations
CAGR (%)	52.81	31.29	65.86	38.24
Sharpe	1.72	1.24	1.46	1.12
Sortino	3.00	2.04	2.68	1.98
Max drawdown (%)	23.59	25.79	24.51	32.17
Calmar (CAGR / max DD)	2.24	1.24	2.69	1.22
Total return (%)	733.0	290.0	1,154.6	404.9
Profit factor	6.89	3.42	1.93	1.71
Win rate (%)	82.8	74.5	57.4	58.0
Closed trades	64	68	122	118
Avg holding (days)	8.5	8.1	5.3	4.8
Quarter wins (of 20)	11	10	10	10
Time invested (%)	29.9	29.8	35.5	32.2

Table 6.1 – Median metrics of parameter variations (long + short, 1 Jan 2021 – 31 Dec 2025, costs and carry deducted). Final = the frozen strategy in Section 3. Calmar and profit factor (gross profit / gross loss of net episodes) are derived from the reported figures.

- All 100 variations are profitable for both assets.
- The final settings sit near the top of the neighbourhood. Few variations match their CAGR, and none beat their drawdown on ETH. This is expected, because the parameters were chosen on this same history, so the final figures are best read as an upper estimate.
- BTC holds up better than ETH. ETH is more parameter-sensitive, with a wide spread of CAGR across variations.
- Quarter wins are 10 of 20 at the median for both assets, so the "more than half of quarters" target depends on the exact parameters.

7. Required Performance Statistics

Separate 100,000-USDT accounts, 1 Jan 2021 – 31 Dec 2025. L/S = long + short; L = long only. Costs and carry are already deducted.

Metric	BTC L/S	BTC L	ETH L/S	ETH L
Final equity	832,999.45	575,382.29	1,254,581.90	587,590.48
Net profit	732,999.45	475,382.29	1,154,581.90	487,590.48
Total return (%)	733.00	475.38	1,154.58	487.59
CAGR (%)	52.81	41.91	65.86	42.51
Gross profit (winning net episodes)	857,420.18	553,420.91	2,395,321.27	1,060,011.05
Gross loss (losing net episodes)	124,420.73	78,038.61	1,240,739.36	572,420.57
Gross profit before fees/carry	916,069.75	588,213.52	2,495,151.74	1,085,574.85
Gross loss before fees/carry	116,161.64	73,093.43	1,151,382.04	530,021.79
Total closed trades	64	45	122	65
Executed orders	128	90	244	130
Winning trades (%)	82.81	86.67	57.38	43.08
Average winning trade	16,177.74	14,190.28	34,218.88	37,857.54
Average losing trade	-11,310.98	-13,006.44	-23,860.37	-15,470.83
Largest winning trade	142,131.60	118,350.85	318,840.32	157,472.85
Largest losing trade	-35,121.44	-26,610.01	-74,994.72	-37,039.33
Maximum drawdown (%)	23.59	23.59	24.51	28.69
Sharpe	1.72	1.54	1.46	1.16
Sortino	3.00	2.63	2.68	2.15
Avg holding (days)	8.52	10.39	5.32	7.00
Max holding (days)	60.00	60.00	41.21	41.21
Buy-and-hold total return (%)	202.25	202.25	302.54	302.54
Fees / slippage paid	64,144.56	39,737.80	179,571.58	67,962.58
Short carry paid	2,764.10	0.00	9,616.22	0.00
Mean gross exposure (%)	26.83	24.10	31.15	23.71
Time invested (%)	29.86	25.60	35.53	24.93
Longest underwater (days)	298.67	372.74	295.32	419.05
Long episode net P&L	586,583.06	475,382.29	784,132.78	487,590.48
Short episode net P&L	146,416.39	0.00	370,449.12	0.00

8. Final Assessment and Limits

- BTC: 832,999 USDT, 52.81% CAGR, Sharpe 1.72, max drawdown 23.59%, beat buy-and-hold in 11 of 20 quarters.
- ETH: 1,254,582 USDT, 65.86% CAGR, Sharpe 1.46, max drawdown 24.51%, beat buy-and-hold in 10 of 20 quarters. Higher return, but less consistent.
- Removing shorts cuts CAGR to 41.91% (BTC) and 42.51% (ETH).
- Parameter sensitivity: all 100 nearby variations are profitable for both assets. The final settings were chosen on this history and sit near the top of that range.

- Limits: both accounts are flat much of the time, with underwater spells near 300 days. Costs are a fixed assumption, not an order-book model. Short results assume borrow and carry, not proven availability. Hidden-segment performance is not established, and synthetic tests are stress cases, not proof.

Reproducibility: source-file SHA-256 prefixes: BTC fa10b463b83a16aa; ETH 4a776f466b788222. Code: bitcoin_crash_strategy.py, eth_final_strategy.py, sweep.py.

9. References

BTC

1. Garcia & Schweitzer (2015), Social signals and algorithmic trading of Bitcoin — <https://pmc.ncbi.nlm.nih.gov/articles/PMC4593685/>
2. Otabek & Choi (2024), Multi-level deep Q-networks for Bitcoin trading strategies — <https://www.nature.com/articles/s41598-024-51408-w>

ETH

3. Yeganeh et al. (2025), Using the attention layer mechanism in construction of a novel ratio control chart: An application to Ethereum price prediction and automated trading strategy — <https://www.sciencedirect.com/science/article/pii/S0952197624018104>
4. Sebastião & Godinho (2021), Forecasting and trading cryptocurrencies with machine learning under changing market conditions — <https://link.springer.com/article/10.1186/s40854-020-00217-x>

10. Code Repository

GitHub: <https://github.com/vaiidiik/inter-iit-bootcamp-quant>